customer plans, troubleshooting, processing payments, answering queries, and so on. AI will also be used at the wireless network layer to improve efficiency and optimise performance, study network traffic and feedback from sites to predict and avoid points of failure, and reduce power consumption.

Intelligence in vehicles: It is not like we are going to see self-driving cars on the roads, but we are going to see improved cognisance and response in mainstream vehicles, thanks to the influence of AI in their advanced driver assistance systems (ADAS). In EVs especially, we can see improvements in energy optimisation, battery management, predictive maintenance, and recharge suggestions based on battery status and charging station proximity. Vehicle makers also use AI, VR, and simulations for advanced testing to improve safety.

Blockchain's renaissance: After the 2022 crash, blockchain and cryptocurrency seem to have come back in 2024, thanks to significant institutional capital inflows and more effective regulations. And the comeback has more to do with blockchain's core strengths than cryptocurrency's charm! Digital identity management, decentralisation, and peer-to-peer settlements, which form the crux of blockchain technology, have found good use in many enterprise applications, ranging from supply chain management and customer engagement, to secure voting and fraud prevention.

Significant improvements in blockchain technology, such as scalability, interoperability, carbon neutrality, and AI integration, along with regulated stablecoins, have led to greater enterprise adoption of blockchain tech. On one hand, we see companies like Starbucks using blockchain tech for customer engagement. In contrast, on the other, we see traditional financial institutions integrating blockchain tech to eliminate intermediaries and automate manual processes securely. Blockchain platforms like JP Morgan's Kinexys (earlier Onyx) support the tokenisation of assets, secure information transfer across financial institutions, and instant cross-border money transfer. Tech majors like Microsoft and IBM also offer enterprise-grade blockchain tools and platforms. Analysts feel that blockchain technology and its adoption will continue to grow in 2025, albeit with a few challenges like price volatility, scalability bottlenecks, and security risks like quantum computing decryption.

Electronics manufacturing in India: The recently held CII Electronics Summit highlighted significant opportunities for the electronics and semiconductor industry in India, particularly in components manufacturing, automotive electronics, and emerging technologies. Growing domestic demand, together with government policies in favour of self-reliance (Atmanirbhar Bharat), is all set to push the sector into a sustainable growth trajectory.

In a July 2024 report, NITI Aayog projected that in a business-as-usual scenario, India's electronics manufacturing could escalate to $278 billion by FY30, with $253 billion from finished goods and $25 billion from components manufacturing. However, the report recommended that with proper fiscal and policy support, India should aim to achieve $500 billion in electronics manufacturing by FY30, comprising $350 billion from finished goods manufacturing and $150 billion from components manufacturing.

Apple's iPhone production in India reached a freight-on-board (FOB) value of $10 billion in the first seven months of FY25 (April-October 2024), with Tata Electronics not only assembling iPhones but also supplying components to Apple. Tata Electronics is looking to diversify its electronics manufacturing business further and is in talks with industry majors like Microsoft, Dell, and HP.

Encouraged by Tata's success, 25 large industrial houses, including TII Murugappa Group, Pidilite, GMR Group, RP Sanjiv Goenka Group, and Bajaj Hindustan, are now working with the government to enter the global electronics supply chain, with components, sub-assemblies, and raw materials—according to the Indian Cellular and Electronics Association (ICEA). However, according to the NITI Aayog report, India currently contributes only 1% of the $4.3 trillion global electronics manufacturing sector. So, we need to get the ball rolling real fast to achieve the ambitious target of $500 billion by 2030.

Each year begins with a million dreams, evolves into a billion achievements, and leaves us aiming for the trillion. As the cycle renews with the arrival of a new year, so does our hope and determination. Stay tuned as we bring you updates on developments in the months to come. END

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